

FPGA Offloading Survey for TRACCC

Weekly Progress Report

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Master's Thesis Progress Report

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Outline

- 1 Executive Summary
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- 3 Profiling Results
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Survey Objective

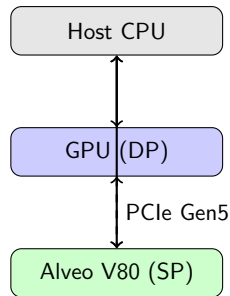
Goal: Evaluate hybrid GPU/FPGA execution for TRACCC particle tracking

Target Platform:

- AMD Alveo V80 accelerator
- 10,848 DSP58 slices (native FP32)
- 32 GB HBM2e (819 GB/s)
- PCIe Gen5 x8x8 (~ 64 GB/s)

Survey Scope:

- Precision analysis (FP32 vs FP64)
- NCU/Nsys profiling
- FPGA suitability assessment



Key Question:

Can FPGA offload latency-bound GPU kernels?

Key Finding: FPGA Offload Candidate

Primary Finding

propagate_to_next_surface kernel: **63% of GPU time, 93% warp stalls**

Why It's Latency-Bound:

- RK4 integration: sequential MAC operations
- B-field lookup: dependent memory access
- Variable step count: 1–34 steps/track
- 62% wasted GPU cycles due to divergence

Why FPGA is Ideal:

- DSP58: native FP32 MAC chains
- Pipelined execution (no warp sync)
- Dedicated pipelines per track
- Variable steps $\neq$ resource waste



FP32 vs FP64: No Physics Difference

Validation Result:

- χ^2/NDF identical: 1.0042 ± 0.0020
- Pull distributions: σ differs by < 0.003
- All 10,000 tracks tested

Param	FP32 σ	FP64 σ	$\Delta\sigma$
d0	0.987	0.987	0.000
ϕ	0.980	0.980	0.000
q/p	0.980	0.980	0.000
θ	0.953	0.955	0.002
z0	0.987	0.987	0.000

Table: 10 GeV muon pull distributions

Implication for FPGA:

- DSP58 native FP32 $\Rightarrow$ **No precision loss**
- No FP32 $\leftrightarrow$ FP64 conversion needed
- GPU and FPGA produce identical physics

Conclusion

FP32 is physics-equivalent to FP64 for TRACCC Kalman filtering. DSP58 native FP32 is safe for FPGA offloading.

GPU Kernel Time Breakdown

Kernel	Time	Target
propagate_to_next_surface	63.0%	FPGA
build_tracks	14.6%	GPU
apply_interaction	6.3%	FPGA
find_tracks	4.2%	FPGA
make_doublets	3.9%	FPGA
select_seeds	3.1%	FPGA
form_spacepoints	2.6%	FPGA
CCL clustering	1.8%	FPGA
FPGA suitable	80.5%	

Propagation Bottleneck:

- 93% warp stall cycles
- Register pressure: 96–128 regs
- L1 cache hit: 46–54%
- Latency-bound, not compute-bound

Why GPU Struggles:

- SIMT model penalizes divergence
- RK4 is inherently sequential
- Memory access is unpredictable

Proposed GPU/FPGA Partitioning

Alveo V80 (Single Precision):

- RK4 propagation (63% GPU time)
- B-field polynomial evaluation
- Chi^2 computation + threshold
- Measurement matching
- CCL clustering
- Seeding/doublets

V80 Resources:

- 10,848 DSP58 $\Rightarrow$ ~ 98 track pipelines
- 32 GB HBM2e for geometry
- 190W TDP

GPU (Double Precision):

- Covariance matrix updates
- 6×6 matrix inversions
- Chi^2 accumulation
- Track deduplication
- Final track fitting

Expected Benefits:

Metric	GPU	Hybrid
Warp stalls	93%	0% (FPGA)
Throughput	1 $\times$	1.5–2 $\times$
Power	300W	490W total

CRITICAL: Per-Step Sync Barrier

- CKF requires GPU $\leftrightarrow$ FPGA sync per step
- 15 surfaces $\times$ sync overhead
- If overhead $>200\mu\text{s}/\text{step}$: **not viable**
- **Must validate before development**

PCIe Latency Analysis:

Scenario	Overhead	Status
Params only	2.7%	✓OK
Full state	13%	Concerning
+ Jacobians	18%	✗Bad

Other Risks:

- DSP58 resource estimates: theoretical
- XRT integration complexity
- Vitis HLS learning curve
- B-field too large for BRAM (139 MB)

Mitigations:

- Cache covariance in HBM2e
- Transfer params only (24 bytes/track)
- Use V80 HBM for B-field
- Start with propagation kernel only

Implementation Roadmap

Phase 1: Validation (Current)

- ✓ FP32 vs FP64 precision validation
- ✓ NCU/Nsys profiling analysis
- ✓ FPGA suitability assessment
- ✗ Sync barrier overhead measurement

Phase 2: Prototype

- RK4 propagation kernel in Vitis HLS
- Single-precision validation vs GPU baseline
- XRT host API integration

Phase 3: Integration

- GPU↔FPGA data path
- Full CKF loop with hybrid execution
- Performance benchmarking

Immediate Next Step

Key Findings:

- 1 propagate_to_next_surface: 63% of GPU time, 93% stalls — ideal FPGA candidate
- 2 FP32 $\equiv$ FP64 for physics — DSP58 safe
- 3 80% of GPU work is FPGA-suitable
- 4 V80 can support $\sim$ 98 parallel track pipelines

Open Questions:

- 1 Per-step sync overhead (critical blocker)
- 2 Actual DSP58 utilization
- 3 XRT integration complexity

Recommendation:

Proceed with sync overhead validation before any FPGA development.

Survey Document

Full analysis: doc/survey-fpga.md (3000+ lines)